Reconstructing 3D objects from fMRI

Data and inference contract

Provided data only: 533 training scans / 292 objects, 125 validation scans / 67 objects, 167 blind test scans. Each scan has 10 time points and 31,141 voxels. Predictions contain 2,048 surface points in the supplied canonical frame (origin centred, z up). Test inference reads fMRI only; subject IDs are available but unused by the selected model. No pretrained models or external object models are used.

Approach

Temporal features: delta. delta means the last-five-time-point mean minus the first-five mean; mean_delta concatenates temporal mean and delta. Spatial standardization is performed separately for each scan. Keep 31141 highest-variance features, then train-fitted voxel scaling. Brain PCA components: 0 (0 means no PCA). For each unique training object, downsample all 32 binary silhouettes from 64 x 64 to 8 x 8, then fit 32 shape PCs. Weighted Ridge (alpha 1000) predicts shape descriptors; each object has equal total regression weight. Retrieve the nearest training descriptor using cosine distance and copy its 3D points unchanged. Training images and voxel-map coordinates are not used.

Selection and leakage prevention

Five GroupKFold splits on training object IDs. Feature selection, scaling, brain PCA, shape PCA, Ridge and retrieval bank are fitted using each training fold only. A target-only pairwise shape cache accelerates scoring and never enters fitting. Select highest grouped-CV identification, with Chamfer as tie-breaker. Candidate additions were motivated by three-fold training category experiments; validation is reported after grouped selection. No hidden test targets are available.

Candidate	Features	Brain PCs	Alpha	Retrieval	CV Chamfer	CV ident.
1	mean	32	10	euclidean	0.1658	51.04%
2	mean	64	100	euclidean	0.1682	52.36%
3	mean_delta	64	1000	euclidean	0.1638	53.16%
4	delta	0	100000	euclidean	0.1565	53.74%
5	delta	0	100000	cosine	0.1775	62.04%
6	delta	0	1000	cosine	0.1764	62.94%

Selected candidate 6; seed 42. CV identification uses only fold opponent objects.

Validation results, uncertainty and controls

Method	Chamfer	F@0.05	IoU	SSIM	Ident. %	Composite
Generic medoid	0.1562	0.3947	0.4384	0.6919	48.7636	-0.0128
Category retrieval	0.1837	0.3527	0.4005	0.6632	52.8485	-0.0552
Selected model	0.1763	0.3755	0.4271	0.6686	60.3030	0.0519

Metrics from supplied evaluate.py; Chamfer lower is better, other metrics higher. Composite is unclipped.

Metric	Mean	95% group-bootstrap interval
chamfer	0.1763	[0.1612, 0.1921]
fscore	0.3755	[0.3324, 0.4227]
silhouette_iou	0.4271	[0.3905, 0.4687]
silhouette_ssim	0.6686	[0.6459, 0.6915]
identification	60.3030	[54.7818, 65.9097]
composite	0.0519	[-0.0529, 0.1566]

Uncertainty scope

2000 percentile-bootstrap replicates resample 67 whole object groups, retaining all scans from each selected group. Identification retains the original opponent set; composite uses fixed generic/oracle anchors. These are conditional intervals, not refitted-model intervals or the hidden gate's test.

Global fMRI pairing control	Identification	Composite
Shuffle 0	49.61%	-0.1683
Shuffle 1	46.67%	-0.2185
Shuffle 2	51.08%	-0.1288

Brain-use evidence

Correct pairing identification: 60.30%. Within-subject pairing permutation test (2000 permutations): null mean 49.95%, one-sided p=0.0004998. Brain dependence is supported by these validation checks. The supplied evaluator and checker do not implement the hidden significance gate; hidden-test passage cannot be guaranteed.

Examples, limitations and reproducibility

Ground truth / val_0089



Prediction / val_0089

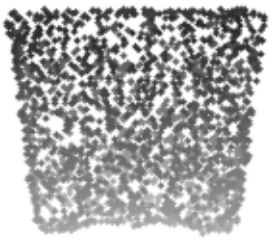


Best validation Chamfer: 0.0286; examples selected by observed error.

Ground truth / val_0093



Prediction / val_0093



Worst validation Chamfer: 0.3961; examples selected by observed error.

Limitations and what did / did not work

Retrieval preserves realistic canonical geometry but cannot generate new shapes or concavities. The original high-variance PCA baseline had 49.43% validation identification and composite -0.0396. Training category experiments favored temporal delta features; grouped CV determined the final choice. A constant training medoid can score reasonably on geometry while carrying no scan information. Subjects have unequal sample sizes. Global shuffled controls may mix subjects; the separate pairing test preserves subjects. Geometry remains weaker than the medoid; the composite interval spans zero.

Reproduction and runtime

Python 3.12.13, CPU only, 4 BLAS threads; 16 logical CPUs available. Recorded completed stage time: 242.6 seconds (CV reused). Run: python run_project.py; to reuse the selected config: python run_project.py --reuse-config. See code/steps.md for environment setup, stage commands, expected runtime, outputs and checks. Exact dependency versions: code/requirements.lock.txt; packaged timings: code/evidence/runtime.json. The supplied data must be placed in code/data when running the packaged source.

Citation and licence

Gao et al., MinD-3D (arXiv:2312.07485) and MinD-3D++ (arXiv:2409.11315). The PDF identifies fMRI-Shape as Apache-2.0. LICENSE_fMRI-Shape.txt was not present in this workspace; retain the original licence from the provided archive if available.